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TWO PHASES OF CRETACEOUS—TERTIARY MAGMATISM IN THE EASTERN DESERT OF EGYPT: PALEOMAGNETIC, CHEMICAL AND K-Ar EVIDENCE

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ABSTRACT

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Paleomagnetic and radiometric data confirm the existence of two phases of igneous activity in the late Mesozoic and Cenozoic history of eastern Egypt. In the mid to late Cretaceous, the earlier and more extensive episode produced the Wadi Natash alkaline basalts and numerous small trachytic intrusions. The ring complexes of Abu Khrug and Gebel El Kahfa were also emplaced at this time. The mean paleopole position for the intrusive rocks is at 65.3°N 111.2°W, A_{95} = 11.4 (31 sites). Chemical analyses of the trachytes show that they are enriched in Nb, Zr and Y and depleted in Sr, Ba, Cr and Ni suggesting either prolonged fractionation or limited partial fusion.

The second phase of activity, during Oligo—Miocene time, is represented by small tholeiitic basalt flows on the Red Sea coast and along the Nile valley. The mean paleopole position for the flows is at 68.2°N 101.5°E, A_{95} = 27.8 (3 sites). The wide scatter is interpreted as reflecting secular variation.

The chronological and chemical data are used with studies of sedimentary formations coeval with the igneous rocks to apply a general model of continental rifting to the northern Red Sea. The earlier, alkaline phase of igneous activity is related to regional uplift while the second, tholeiitic phase is associated with widening of the rift valley.

I. INTRODUCTION

Scattered throughout the Eastern Desert of Egypt and at certain localities in the Western Desert are outcrops of igneous rocks which post-date the late Precambrian to early Phanerozoic Pan-African granites. Rarely do stratigraphic conditions permit a close definition of the age of the extrusive rocks, as for example, by the association of the Wadi Natash Volcanics with the

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Nubian Sandstone or in the case of basalts inter-bedded with Neogene sediments at Gebel Qatrani. With relatively few radiometric dates available, most workers have had to rely upon field criteria and composition to attempt a classification of the intrusions. Commonly these were grouped, from oldest to youngest, as acidic, intermediate, basic and alkaline intrusions. In the course of three field seasons we have collected samples from some 200 sites in the igneous rocks. The results to date suggest that the majority of these, including most of the acidic, intermediate and basic intrusions, are associated with the closing stages of the Pan-African event (Davies et al., 1980). The remaining, younger rocks include some which were emplaced since the mid to late Cretaceous and which constitute the subject of this paper.

Igneous activity since the mid to late Cretaceous occurred in two district phases which we relate to the tectonic history of the northern Red Sea. The earlier and more extensive phase consists of alkaline to peralkaline rocks emplaced during the late Cretaceous. The younger phase produced tholeiitic basalts, mostly during the Oligo-Miocene. The ages and compositions of these rocks, combined with studies of coeval sediments, provide a test of some aspects of a model proposed by Falvey (1974) for the development of passive (Atlantic-type) continental margins. In this model, continental break-up is preceded by regional updoming, alkaline magmatism and shedding of detritus from the locus of uplift. The uplifted area subsequently undergoes block faulting and a rift valley is formed. Continued upward heat flow beneath the rift axis causes deep crustal metamorphism, density increase and column reduction. This in turn leads to a widening of the rift valley by basin collapse, and to transport of sediment towards the axial valley. Approximately 50 m.y. after the initial uplift, continental separation accompanied by tholeiitic volcanism begins. This takes no account of the subsequent spreading behaviour, that is, it is independent of the two spreading stages, proposed and discussed by Girdler and Styles (1976), Davidson and Rex (1980) and Richardson and Harrison (1976).

PALEOMAGNETIC AND RADIOMETRIC STUDY

Alkaline rocks

Regional geology

The late Cretaceous alkaline intrusions of the Eastern Desert may be divided for convenience into three groups. The first, the trachytic (bostonitic) dikes, sills and plugs (B sites) intrude into the sediments of the Hammamet Formation and surrounding metamorphic rocks of the basement complex in the region southwest of Quseir (Fig. 1). Several of these units have been affected by uranium mineralization and these have been studied in detail by Hussein and El Kassas (1971, 1972). In the field the rocks usually form prominent relief and are recognizable by their light colored fine-grained character.

The rocks are composed predominantly of alkali feldspar and subordinate quartz, ferromagnesian minerals, zircon and apatite. In the absence of the radial arrangement of the feldspar laths, considered diagnostic of bostonites, it is suggested here that the rocks be called by the more familiar term, trachyte. El Manharawy (1972) quoted Rb-Sr ages for these rocks ranging from 300 m.y. to 74 m.y. despite the gross similarity of the rocks. Recent K-Ar dates of these rocks, however, give a much closer grouping of ages from 63 m.y. to 92 m.y. (Table I).

The second group of alkaline rocks (NR sites) are located in the vicinity of Wadi Natash. Here lava flows, predominantely alkaline olivine basalt, are interbedded with the upper Cretaceous Nubian Sandstone. The lavas and the

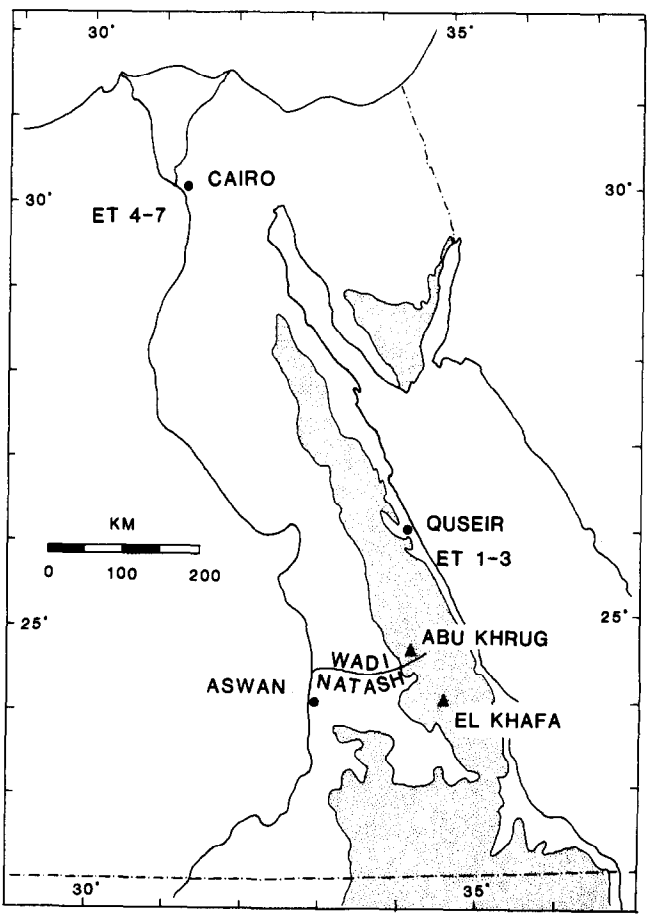


Fig. 1. Sketch map of eastern Egypt, illustrating the location of the sample sites. The B sites are located immediately west of Quseir. The NR sites are at Wadi Natash. The ET sites are Cenozoic volcanics sampled in the Nile Valley and Red Sea areas. The outcrop area of the crystalline basement complex is shown in stippled pattern.

sandstone were studied paleomagnetically by El Shazly and Krs (1973). Both of these units and the adjacent metamorphic basement were intruded by numerous small plugs and ring dikes, mostly of trachytic composition. The trachytes are also believed to be associated with volcanism as one partial ring dike at Geziret Khashim Natash (NR—8) is flanked with volcanic agglomerate.

The third group, the alkaline ring complexes, pose a difficult problem since they formed sporadically in the Eastern Desert over an interval from about 550 m.y. to 80 m.y. (Serencsits et al., 1978; Lutz et al., 1978). We consider here only the youngest of these complexes, the Abu Khrug and the

TABLE I

Summary of radiometric ages of the late Mesozoic—Cenozoic igneous rocks

Unit	Method	Age (m.y.)	Source
A. Quseir region trachytes			
B-4 (EB-74)	K/Ar	92	1
B-3 (EB-57)	K/Ar	91	1
B-3 (EB-59)	K/Ar	90	1
B-7 (EB-96)	K/Ar	88	1
B-4 (EB-70)	K/Ar	87	1
B-11 (EB-128)	K/Ar	68	1
B-15 (EB-151)	K/Ar	65	1
B-14 (EB-145)	K/Ar	63	1
B. Wadi Natash lavas and trachytes			
Alkalic basalt	Rb/Sr	104 ± 7	2
Alkalic basalt	K/Ar	85.6 ± 3	3
Trachyte plug NR-10	K/Ar	86	1
Trachyte plug NR-11	K/Ar	78	4
C. Abu Khrug ring complex			
Biotite	K/Ar	89 ± 2	5
Biotite and whole rock	Rb/Sr	89.5	5
D. El Khafa ring complex			
Biotite and whole rock	Rb/Sr	92 ± 2	5
Biotite and hornblende	K/Ar	91 ± 4	5
Syenites	Rb/Sr	81.4 ± 7	2
E. Tertiary basalts			
El Qubli ET 1	K/Ar	32.4	6
El Bahiry ET 2	K/Ar	25.5	6
Gebel Qatrani ET 4	K/Ar	27 ± 3	7
Gebel Quatrani ET 5	K/Ar	24.7	7

All analyses are whole rock unless otherwise noted. Sources: (1) Ressetar, (1979); (2) Hashad and El Reedy (1979); (3) El Shazly and Krs (1973); (4) Higazy and El Ramly (1960); (5) Serencsits et al. (1978); (6) R. Sedivy (pers. commun.); (7) El Shazly (1977).

Gebel El Kahfa, which overlap in time the period during which the trachytes and Wadi Natash igneous rocks were formed (El Ramly et a., 1969a, b).

Paleomagnetic investigation

The techniques of investigation are standard: an average of six oriented cores drilled at each site, measurements of remanence made using a Digico magnetometer, and progressive demagnetization carried out both thermally and in alternating magnetic fields. One or two pilot samples per site were demagnetized at 50°C or 50 Oe (5 mT) and 100 Oe (10 mT) intervals to determine the optimum temperature or field at which the remainder of the samples were measured as the second of three demagnetization stages. The trachytic rocks (B sites) with few exceptions showed a relatively small drop in magnetic intensity when demagnetized in alternating fields (Fig. 2), but upon thermal demagnetization samples typically retained a single direction of remanence with decreasing intensity (Fig. 3) to temperatures above 600°C. From this it seems apparent that hematite was a principal carrier of remanence, although some titanomagnetite could be present. The presence of hematite was observed in thin section. Site mean directions of magnetization are shown in Fig. 4 with a data summary in Table IIA.

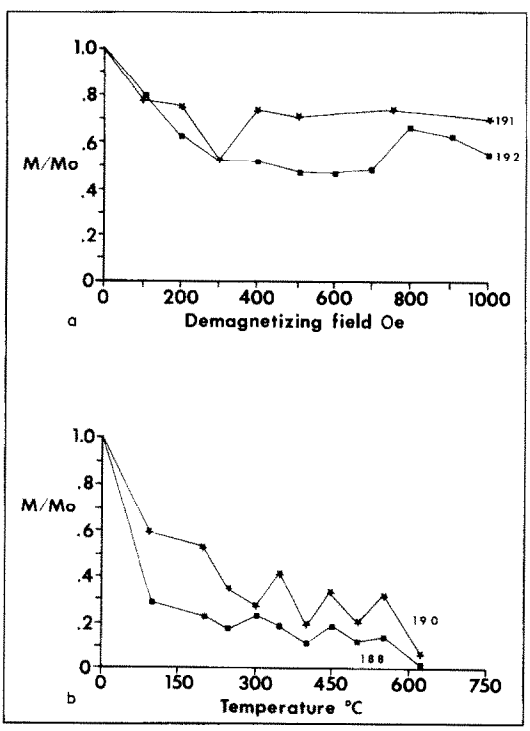


Fig. 2. Normalized alternating field and thermal demagnetization curves of B samples.

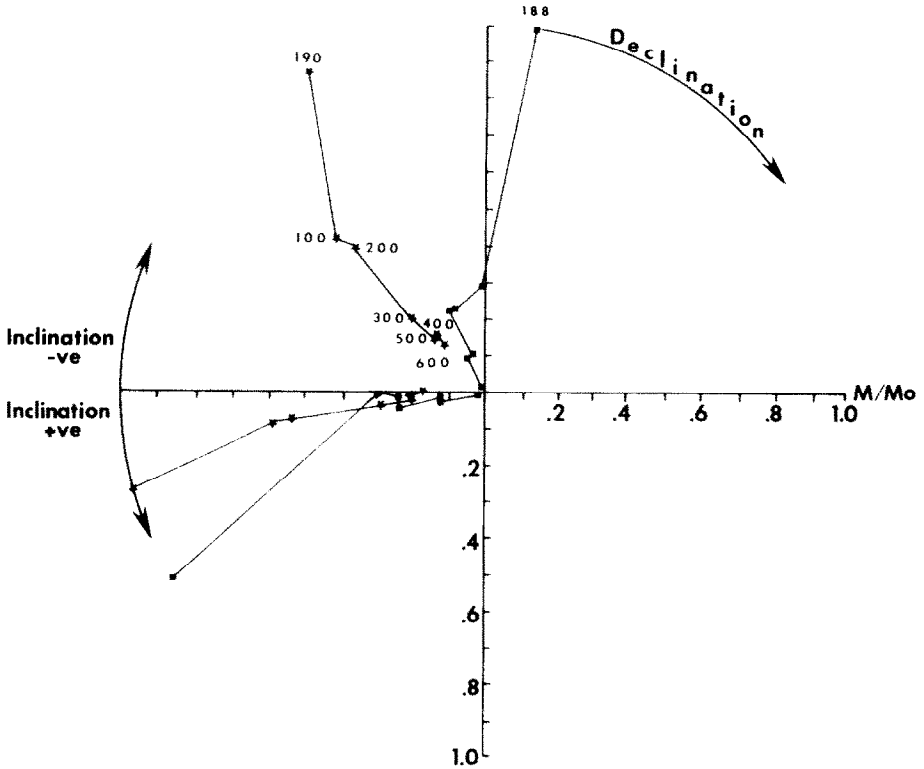


Fig. 3. Normalized stable end point curves from B samples thermally demagnetized at 100°C intervals.

The NR site samples characteristically have a higher NRM (natural remanent magnetization) than the B sites, and alternating field demagnetization suggested that some titanomagnetite was present, even if hematite was the principal remanence carrier (Fig. 5). It is clear from the stable end point diagram (Fig. 6) that both carried the same directions of magnetization. Site mean directions of magnetization are summarized in Table IIB and illustrated in Fig. 7.

The intensity of remanent magnetization of the Abu Khrug and Gebel El Kahfa samples varies over three orders of magnitude, and this together with the high alternating fields used in the cleaning of some rocks suggests that the stronger samples carry a lightning induced IRM (isothermal remanent magnetization). In fact the dispersion found at site AR4, suggests that even at the highest field possible with our apparatus there still may be a residual secondary component. A stable end point diagram (Fig. 8a) illustrates one sample in which fields of 500 Oe (50 mT) were needed for cleaning. Normalized demagnetization curves for the samples in Fig. 8 are given in Fig. 9.

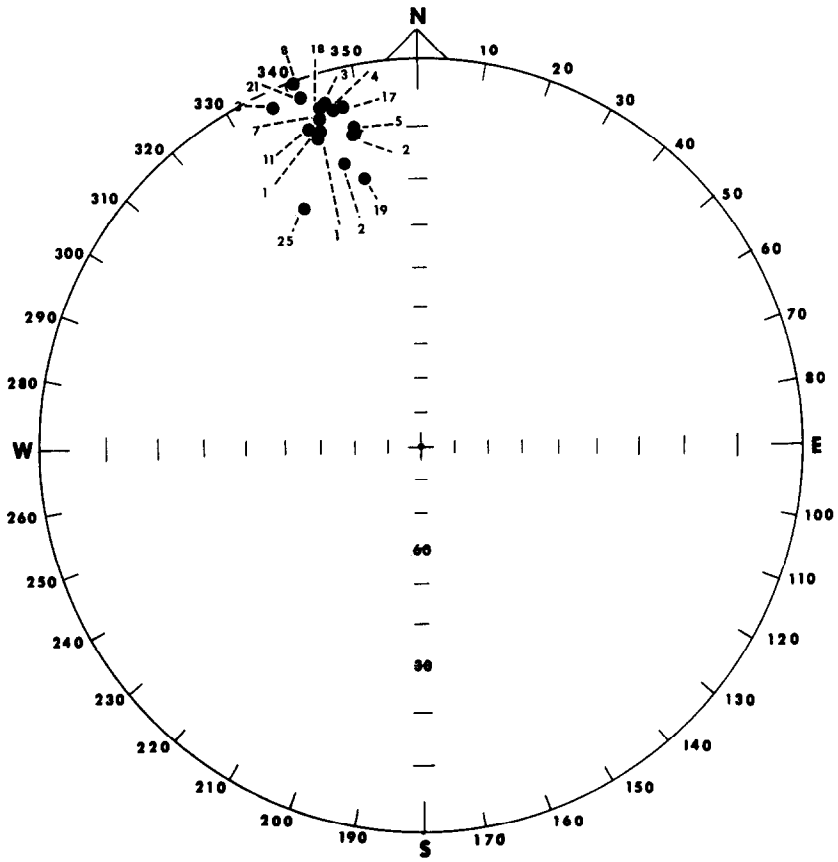


Fig. 4. Stereographic plot of cleaned B site mean directions of magnetization.

The cleaned mean directions of magnetization are shown in Fig. 10, the parameters are listed in Table IIC and D.

The geomagnetic poles calculated for each group of samples is shown in Fig. 11 along with other African Mesozoic geomagnetic poles. Since the African plate was relatively stationary during the Mesozoic, it is not possible to infer a Cretaceous age purely from paleomagnetic considerations. The radiometric ages, however, make it clear that the intrusions were emplaced during the late Cretaceous.

Basaltic rocks

Regional geology

Fairly small basaltic lava flows of Oligo—Miocene age represent the most widespread Cenozoic igneous activity in Egypt (El Shazly, 1977). Although

TABLE II

Summary of paleomagnetic data from late Mesozoic and Cenozoic rocks from the Eastern Desert and the Quatrani Depression

Site	Sampling *a/b*	Declination	Inclination	α_{95}	*K*	*R*	Demagnetization
(A) Trachytes minor intrusions from west of Quseir							
B-1	7/9	343.3	8.0	4.6	173.1	6.965	400°C
B-2	6/7	345.9	15.4	8.3	66.0	5.924	400°C
B-3	9/11	344.7	3.7	4.1	155.3	8.949	500°C
B-4	6/7	346.1	5.4	3.7	318.4	5.984	450°C
B-5	3/4	347.8	9.0	3.4	1243.2	2.998	500°C
B-7	7/8	344.0	6.4	2.5	553.4	6.989	500°C
B-8	5/8	341.0	0.8	1.7	1812.8	4.998	400°C
B-10	7/7	342.0	9.7	3.9	233.9	6.974	400°C
B-11	6/6	340.8	6.0	2.8	558.6	5.991	500°C
B-12	4/5	347.5	9.5	7.7	140.8	3.979	500°C
B-13	4/4	337.2	1.9	4.7	340.9	3.992	400°C
B-17	6/8	347.1	4.5	3.6	345.6	5.986	400°C
B-18	7/7	344.3	3.7	1.8	1030.4	6.994	300°C
B-19	4/5	348.4	19.4	6.6	191.9	3.984	350 Oe
B-21	5/6	342.1	2.2	3.1	597.6	4.993	500°C
B-25	6/8	334.5	21.2	9.1	54.2	5.908	175 Oe
Mean of all sites	16	343.6	7.9	3.1	134.2	15.888	
Paleopole position: 63.0°N 107.7°W.							
(B) Minor intrusions from Wadi Natash							
NR-7	4/6	353.0	34.7	7.1	166.9	3.982	250 Oe
NR-8	4/6	6.2	26.8	26.2	13.1	3.773	350 Oe
NR-9	5/6	18.9	20.5	32.4	6.5	4.387	250 Oe
NR-10	6/6	342.1	2.3	15.7	19.0	5.738	350 Oe
NR-11	5/6	344.0	15.5	10.9	49.4	4.919	350 Oe
Mean of all sites	5	356.5	20.5	18.1	18.1	4.779	
Paleopole position: 75.7°N 131.7°W.							
(C) Sites from the Abu Khrug ring complex							
AR1 Nepheline syenite	4/5	339.7	+6.2	20.8	20	3.853	1,000 Oe
AR2 Essexite gabbro	6/6	331.6	+19.1	11.3	36	5.861	350 Oe

AR3 Nepheline syenite	4/6	339.7	−4.1	14.3	42	3.928	250 Oe
AR4 Essexite gabbro	3/5	337.4	+33.5	42.2	10	2.791	1,000 Oe
AR5 Trachyte porphyry	4/4	337.2	−1.8	16.2	33	3.909	200 Oe
AR6 Trachyte agglomerate	5/6	331.0	−1.9	43.6	4	3.999	750 Oe
Mean of all sites	6	336.1	+8.4	12.9	27.8	5.821	
Paleopole position: 59.1°N 94°W.							
(D) Sites from the E1 Kahfa complex							
KR12 Alkali syenite	4/6	350.2	−6.6	21.4	19	3.845	350 Oe
KR13 Trachyte	4/6	353.6	−9.4	15.3	37	3.919	100 Oe
KR14 Essexite gabbro	3/4	359.1	−0.9	13.0	90	2.978	750 Oe
KR15 Nepheline syenite	2/2	*	*	*	*	*	*
KR16 Trachyte	4/6	340.9	−5.0	22.4	18	3.831	1,000 Oe
Mean of all sites	4	349.1	−5.4	7.5	147.8	3.980	
Paleopole position: 61.1°N 122.4°W.							
(E) Summary of data							
Mean of site means	4	346.1	+7.9	15.5	35.9	3.917	
Mean paleopole position: 4; 65.3°N 111.2°W; A_{95} = 11.4,							
Mean of all sites	31	344.7	+8.2	4.6	32.1	30.066	
Paleopole position: 31; 64.6°N 108.2°W.							
(F) Sites in Egyptian Tertiary basalts							
ET1 Coastal plain Quseir	14/15	196.4	−53.1	3.3	139.2	13.907	200 Oe
ET2 Coastal plain Quseir	5/6	216.7	−29.6	7.0	110.6	4.967	150 Oe
ET3 Coastal plain Quseir	9/9	215.3	−34.4	11.4	21.2	8.623	250 Oe
ET4 Qatrani depression Cairo	9/10	209.3	−59.9	3.4	220.3	8.964	200 Oe
ET5 Qatrani depression	8/11	209.4	−62.5	3.5	241.9	7.971	200 Oe
ET6 Qatrani depression	7/8	204.0	−50.8	1.9	915.9	6.993	200 Oe
ET7 Qatrani depression	8/9	207.5	−58.7	2.6	452.9	7.985	150 Oe
Mean of all sites		209.1	−50.1	10.3	35.2	6.830	
Mean Pole position 3; 68.2°N 101.5°E; A_{95} = 27.8, (ET1; ET2, 3; ET4—7)							

* No consistent data.

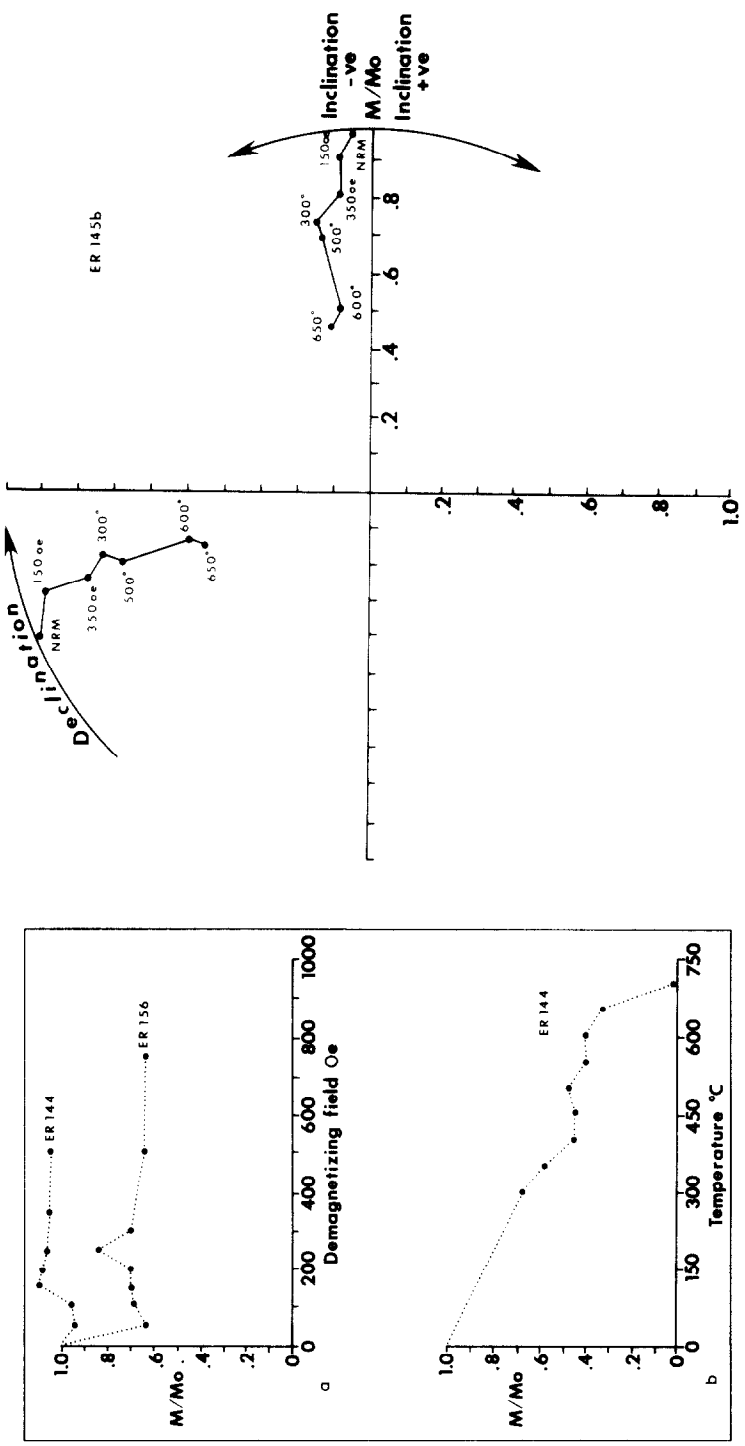


Fig. 5. Normalized alternating field and thermal demagnetization curves of NR samples.

Fig. 6. Normalized stable end point curve from an NR sample.

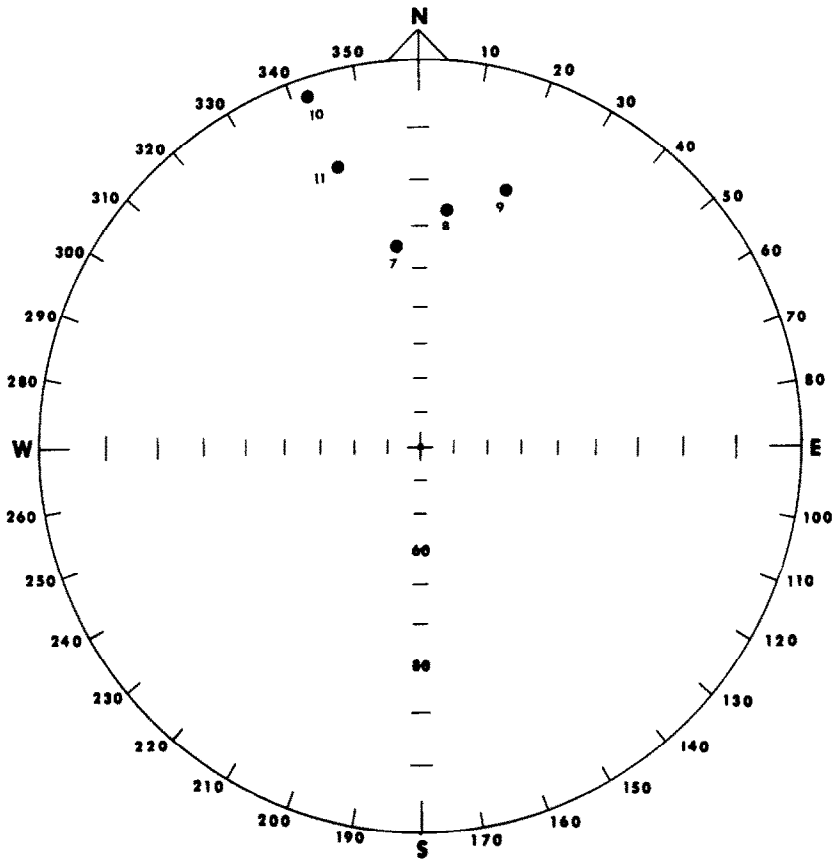
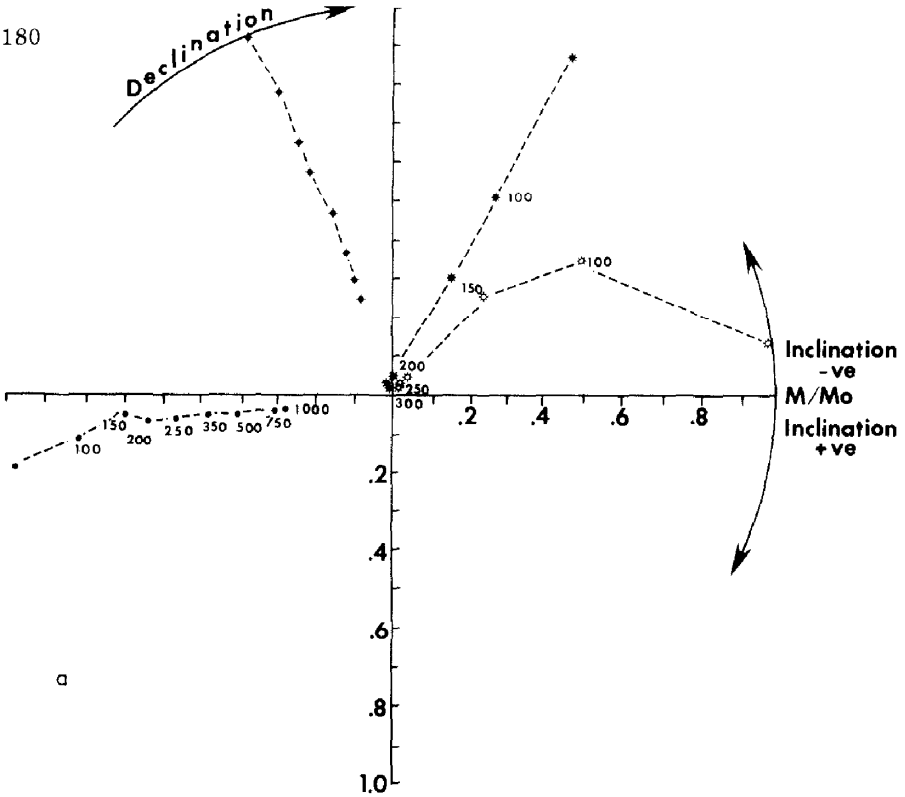


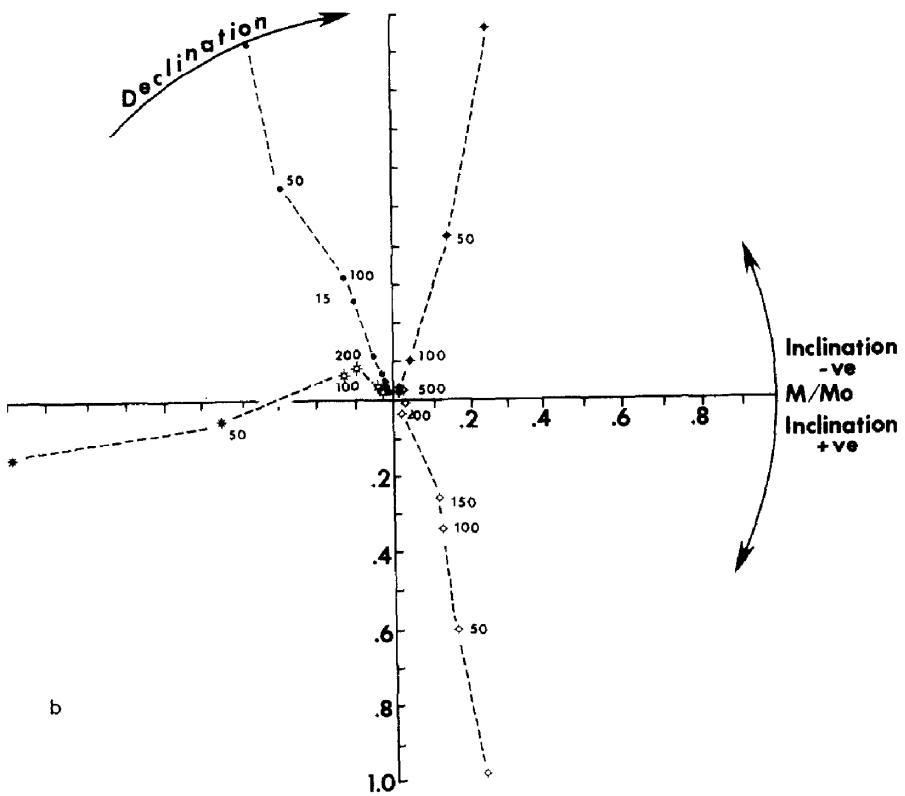
Fig. 7. Stereographic plot of cleaned NR site mean directions of magnetization.

most of the flows are located in the Western Desert and the Nile Valley, a few are in the Red Sea coastal plain. Three of our sample locations (ET 1—3) are along the coastal road south of Quseir. Here, isolated outcrops of tholeiitic basalt approximately 10 m thick are interbedded with Oligo—Miocene (?) sediments. A paleosoil horizon immediately below the basalts shows they were extruded subaerially. Since basalt is not everywhere found above the paleosoil, it appears that the outcrops represent separate flows, rather than a single dismembered unit. Further description of these rocks is given by Sabet (1958).

Our remaining sample sites (ET 4—7) are located in the Qatrani area west of Cairo. At least two flows, reaching 20 m thickness, make up the top of the Oligocene section (El Shazly, 1977). Miocene sands and gravels lie directly above the flows. See Wassif (1977) for additional description of these units.


Declination
Inclination
-ve
M/Mo
Inclination
+ve
100
150
200
250
300
350
500
750
1000
.2
.4
.6
.8
1.0

a


Declination
Inclination
-ve
M/Mo
Inclination
+ve
50
100
150
200
500
.2
.4
.6
.8
1.0

b

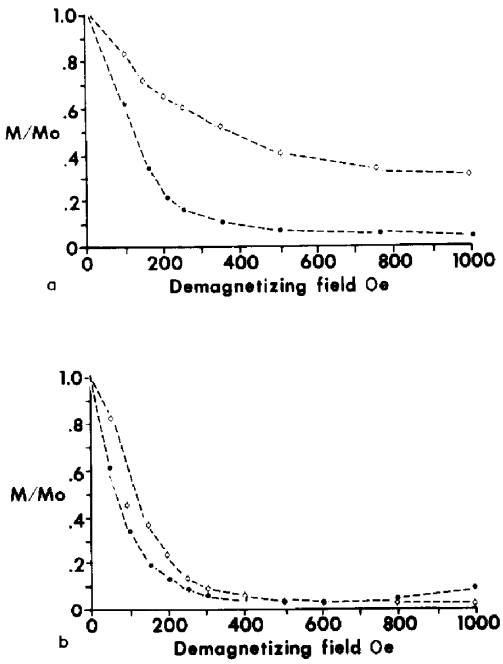
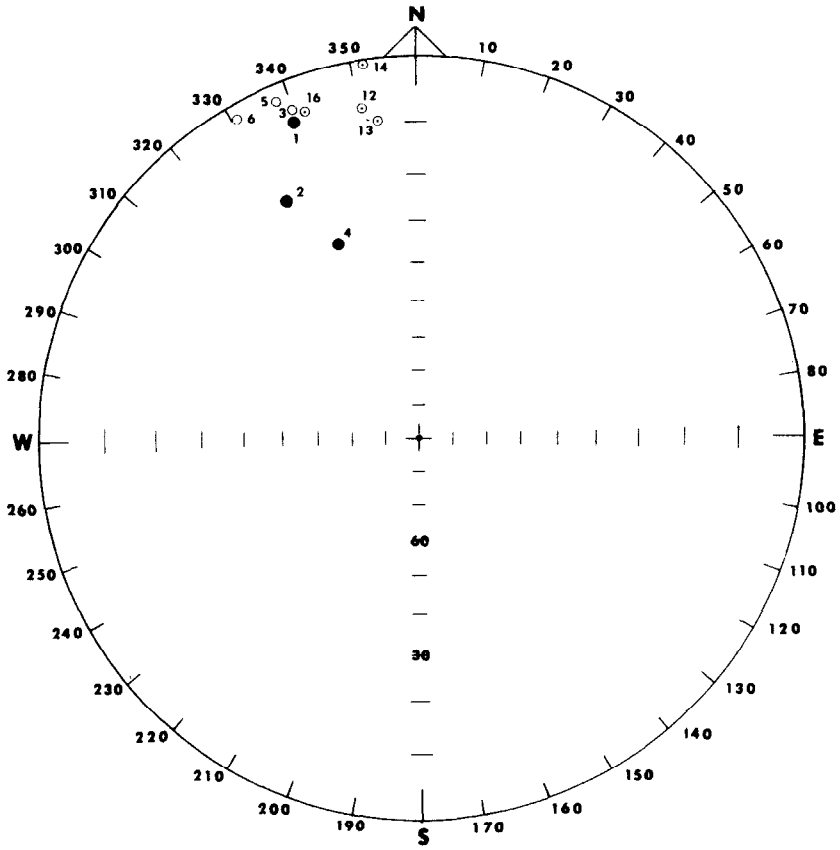


Fig. 9. Normalized demagnetization curves from two sites in (a) Abu Khrug ring complex and (b) Gebel El Kahfa ring complex.

Paleomagnetic and K-Ar results

In spite of the weathered appearance of some of the outcrops, the ET samples invariably had strong, stable magnetizations. Samples responded well to AF demagnetization (Fig. 12) with little change in direction (Fig. 13). Consequently, the NRM directions reported by Wassif (1977) for samples represented by our sites ET 4—7 are statistically indistinguishable from those obtained by us after alternating field treatment. Although the within-site dispersion was low (Table II), there are significant differences between mean directions of magnetization (Fig. 14) and consequently, the paleopoles (Fig. 11). The direction of site ET 1 is well removed from that from ET 2 and 3 while the Qatrani sites give a third direction. Since the units appear undisturbed, the differences in the directions of remanence are regarded as reflecting secular variation between flows extruded an unknown interval of time apart, and are regarded as reinforcing the field interpretation that the out-

Fig. 8. Normalized stable end point curves from two sites in (a) Abu Khrug ring complex and (b) Gebel El Kahfa ring complex.



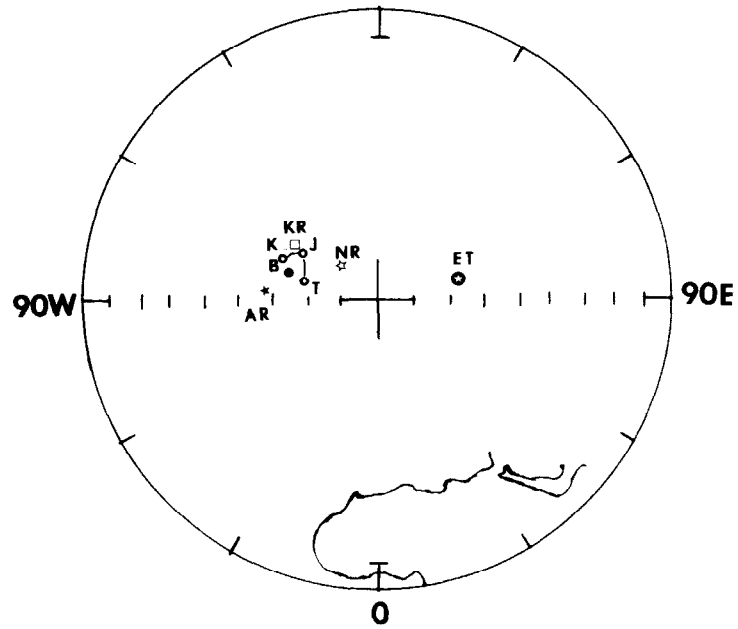


Fig. 10. Site mean directions, after cleaning in alternating fields of samples from two sites in (a) Abu Khrug ring complex and (b) Gebel El Kahfa ring complex.

Fig. 11. Paleopole positions of the Eastern Desert trachytes and of the Oligo-Miocene flows. *B* = trachytes west of Quseir: *NR* = Wadi Natash intrusives; *Ar* = Abu Khrug ring complex; *KR* = El Kahfa complex; *Et* = Egyptian Tertiary flows: *TJK* = mean Triassic, Jurassic and Cretaceous paleopole positions calculated from African data. (Martin et al., 1978.)

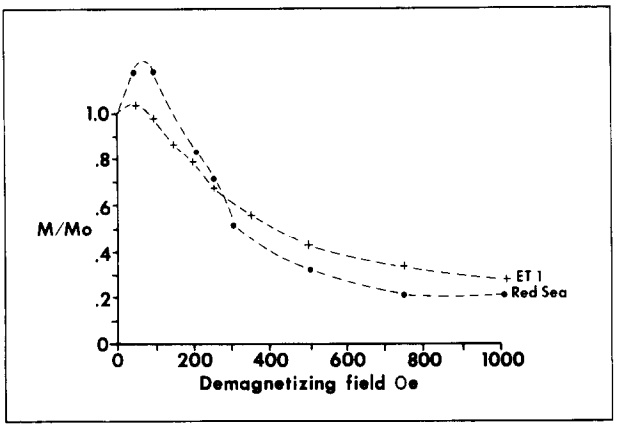


Fig. 12. Normalized alternating field demagnetization curve of an ET sample.

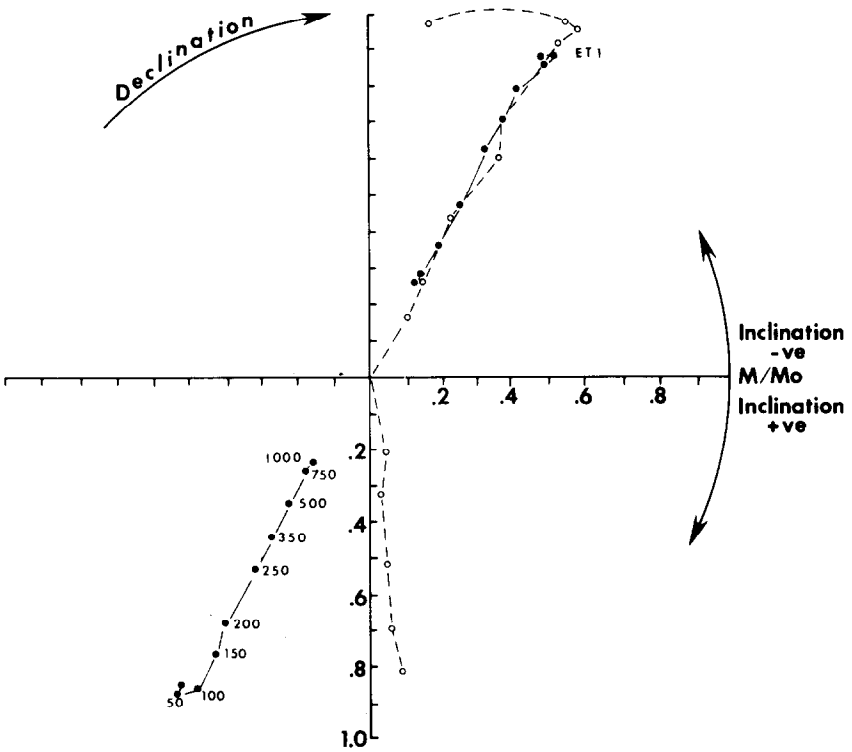


Fig. 13. Normalized stable end curve of an ET sample.

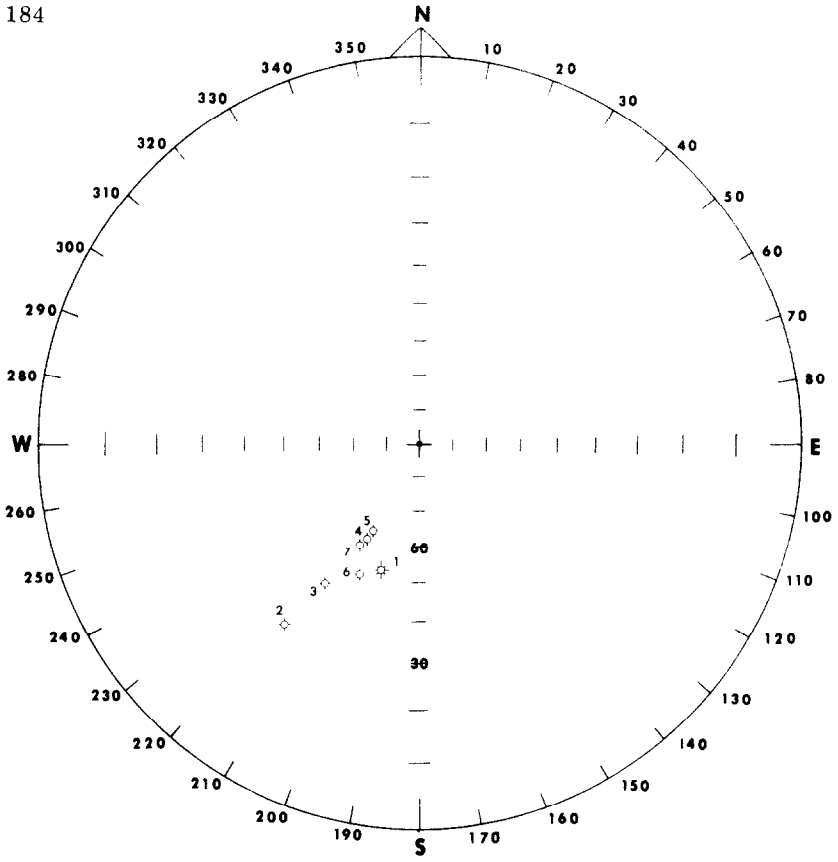


Fig. 14. Stereographic plot of cleaned ET site mean directions of magnetization.

crops do not represent a single, dissected flow. K-Ar ages by El Shazly (1977) and R. Sedivy (pers. comm., 1978) are consistent with the magnetic directions (Table I).

CHEMISTRY OF THE B AND NR SITES

Chemical analyses from six of the B sites and four of the NR sites are presented in Table III, SiO_2 analyses were made colorimetrically. MgO, Na_2O and Al_2O_3 were determined by atomic absorption spectrometry. All other elements were measured by X-ray fluorescence following techniques described by Drez (1977). Iron contents were calculated as total Fe_2O_3. Table III shows a general uniformity of composition, reflecting the petrographic similarities between the units. Total alkalis are high, consistent with the high feldspar contents. About half the rocks are peralkaline with molecular proportions of $Na_2O + K_2O$ exceeding Al_2O_3. Small amounts (< 10%) of quartz, acmite and hypersthene are present in the norm.

TABLE III

Chemical analyses for trachytic intrusions (wt. %)

Sample site	EB57 B3	EB59 B3	EB70 B4	EB74 B4	EB128 B11	EB145 B14	EB151 B15	EB183 B20	EB186 B20
SiO_2	62.15	61.80	63.37	63.47	63.19	61.13	61.64	64.07	64.28
TiO_2	0.24	0.24	0.24	0.25	0.17	0.18	0.17	0.24	0.23
Al_2O_3	14.99	15.08	14.56	14.74	15.11	16.87	16.44	16.09	14.96
Fe_2O_3*	6.44	6.07	5.76	5.97	4.57	4.72	4.51	5.79	5.84
MgO	0.46	0.74	0.47	0.37	0.43	0.39	0.31	0.33	0.20
MnO	0.21	—	0.04	0.18	0.05	0.14	0.19	0.21	0.21
CaO	1.88	1.60	1.90	0.91	1.65	1.36	1.48	1.49	1.00
Na_2O	5.79	5.86	5.38	5.71	3.10	6.82	6.77	6.38	7.16
K_2O	4.33	4.08	4.56	4.17	5.38	4.44	4.58	5.28	5.23
	96.49	95.47	96.28	95.77	93.65	96.05	96.09	99.88	99.11
	(ppm)								
Rb	128	126	145	144	248	103	102	168	161
Sr	13	<5	<5	8	17	11	6	35	47
Y	108	104	168	124	167	127	132	80	76
Nb	208	—	252	252	242	196	232	255	276
Zr	726	785	884	578	1041	807	843	568	759
Cr	38	—	20	9	26	12	37	39	18
Ni	8	—	9	6	8	7	7	10	5
Ba	315	—	11	15	18	19	28	<10	64

TABLE III (cont.)

Sample site	EB209 B24	EB214 B24	ER130 NR8	ER131 NR8	ER135 NR9	ER137 NR9	ER146 NR10	ER149 NR10	ER154 NR11	ER156 NR11
SiO_2	57.24	58.78	63.26	64.19	63.58	64.07	67.64	65.61	65.85	65.93
TiO_2	0.02	0.03	0.38	0.37	0.18	0.19	0.15	0.13	0.17	0.17
Al_2O_3	16.69	17.09	14.70	14.88	15.51	15.84	15.54	15.91	16.02	14.04
Fe_2O_3*	6.36	6.73	6.37	6.59	6.91	6.60	4.69	4.07	5.45	5.40
MgO	0.07	0.03	0.69	0.26	0.38	0.36	0.08	0.09	0.06	0.07
MnO	0.41	0.46	0.21	0.21	0.20	0.14	0.02	0.18	0.17	0.16
CaO	0.71	0.73	1.98	1.95	1.97	1.85	0.62	0.95	1.24	1.19
Na_2O	9.09	9.63	6.09	6.51	5.79	6.03	5.83	6.56	6.68	6.82
K_2O	5.03	4.68	4.21	4.08	5.39	5.16	5.74	4.41	5.20	5.23
	95.62	98.16	97.89	99.04	99.91	100.24	100.31	97.91	100.84	99.01
	(ppm)									
Rb	187	170	81	79	162	158	202	140	124	119
Sr	27	27	257	210	64	42	9	17	28	28
Y	99	101	67	58	126	99	113	107	96	100
Nb	333	370	141	135	244	247	306	297	243	247
Zr	1157	1269	758	754	988	967	771	752	757	710
Cr	14	27	22	20	41	38	24	14	25	17
Ni	10	10	6	5	6	5	8	9	6	6
Ba	<10	98	1160	1056	110	82	78	<10	424	378

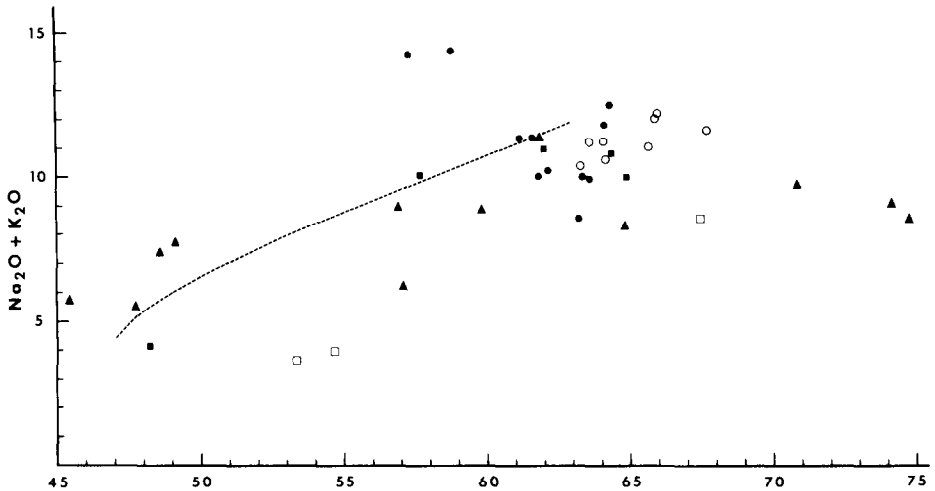


Fig. 15. Weight percent silica versus total alkalis. Key to symbols: solid circles-B sites, open circles-NR sites, triangles-Wadi Natash lavas (from Sayyah and El Shatoury, 1973), open squares- and closed squares-Wadi Masour lavas and NR sites, respectively (from El Ramly et al., 1971). The dashed line is the general trend of Hawaiian alkaline rocks (after Cox et al., 1970).

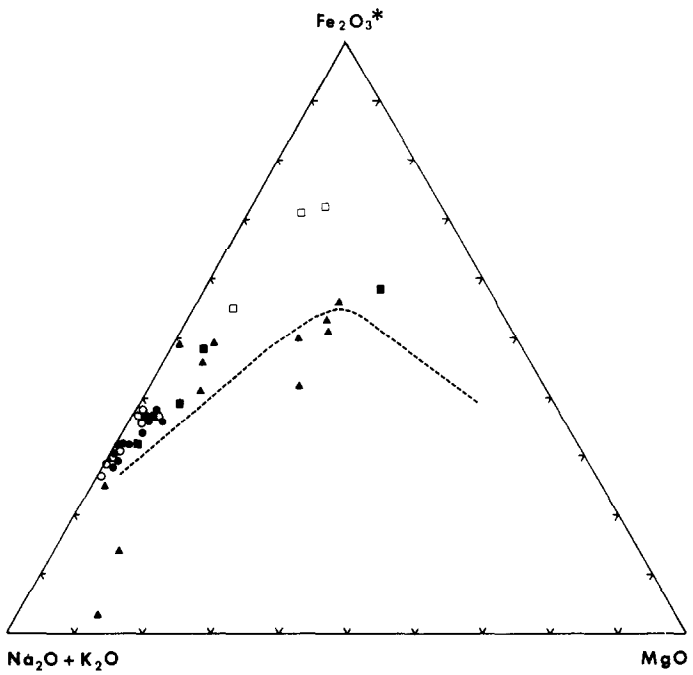


Fig. 16. AFM diagram of the trachytes and the Wadi Natash region lavas. Symbols are the same as in Fig. 15. The dashed line is the general trend of Hawaiian alkaline rocks (after Cox et al., 1970).

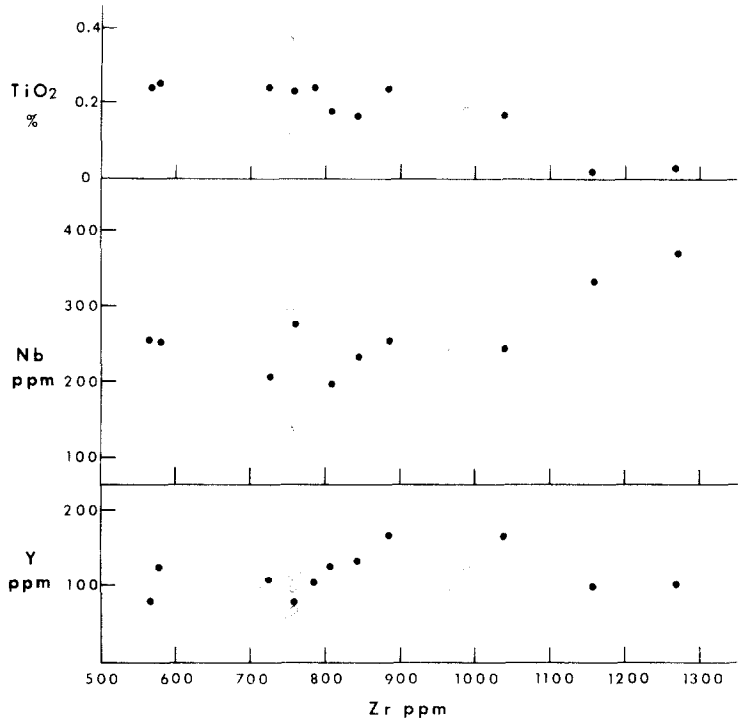


Fig. 17. Plot of TiO_2, Nb and Y versus Zr from the new analyses of the trachytic intrusions. Symbols are the same as in Fig. 15.

Major element variations, shown in an alkali against silica diagram (Fig. 15) and on an AFM diagram (Fig. 16), are compared with analyses taken from the literature for the Wadi Natash lavas (Sayyah and El Shatoury, 1973) and the nearby Wadi Masour lavas (El Ramly et al., 1971) and the trend of the Hawaiian alkaline series (as shown in Cox et al., 1970). In Fig. 15, the Egyptian rocks show a slightly less alkaline character than the Hawaiian rocks, with rhyolitic compositions at high SiO_2 contents. Any trend among the Egyptian rocks is only poorly defined. Fig. 16 shows an absence of Mg-rich compositions but we cannot determine if this is a reflection of sampling. A trend is somewhat more evident in Fig. 16 although these is a gap between the more Fe-rich Wadi Natash lavas and other analyses.

The trace element contents are characterized by high abundances of Nb, Zr and Y and by low levels of Sr, Ba, Cr and Ni. Berlin and Henderson (1969) attributed similar low Sr and Ba contents to extended crystallization of feldspar during fractionation. Weaver et al. (1972) argued for a fractional crystallization origin for a similar group of trachytes by correlating various incompatible elements, especially Nb and Zr. Our data, however, fail to show any discernible trend with Zr variation (Fig. 17).

DISCUSSION

The paleomagnetic and radiometric data presented here provide a means of linking the igneous events to the tectonic history of the northwestern margin of the Red Sea. As indicated earlier there are two phases of igneous activity, the first being volumetrically much more extensive than the second. The opening of the earlier phase was marked by the extrusion of alkaline basalts of which the Wadi Natash lavas are the best known (El Shazly and Krs, 1973), although slightly older lavas are reported from the Wadi Araba in the northern Eastern Desert. The paleopole position of the lavas is indistinguishable from those obtained from the intrusions reported here. The radiometric age dates and the meagre fauna from the basal Nubian Sandstone Formation (Ward and McDonald, 1979) are consistent with a late Cretaceous age for the igneous units. The paleopole positions compare closely with those from other African rocks of similar age, on the assumption of zero relative motion between the Nubian and other African plates (Fig. 11).

The second phase of magmatic activity was neither as voluminous nor as compositionally diverse as the first phase. The peak of activity occurred during the Oligo—Miocene with the extrusion of minor tholeiitic basalts along the Red Sea coast and in the Nile Valley although the radiometric ages suggest a wholly Oligocene age. The scattered directions reflect the small number of sites and hence the mean direction may result from an inadequate sampling of secular variation.

An association of alkaline magmatism, continental arching and rifting has been noted on numerous occasions (e.g. Bailey, 1964; Le Bas, 1971). Other rifted continental margins display an abundance of tholeiitic material (e.g. the Triassic igneous province of eastern North America). Falvey (1974) proposed that these types of igneous activity occur in a predictable sequence and are of limited duration. The ultimate mechanism is an upwelling thermal anomaly which results in (1) doming, extensive surficial erosion, and alkaline magmatism; and (2) metamorphism of the lower crust, attendant density increase and subsidence of an axial valley, transport of sediments into the axial valley, and eventual tholeiitic magmatism as crustal separation begins.

The sequence and timing of the Cretaceous and younger igneous and tectonic events found in the northern part of the Red Sea show a remarkable concordance with Falvey's predictions. Doming of the Nubian, shield occurred during the mid to late Cretaceous (Swartz and Arden, 1960; Garson and Krs, 1976). Paleocurrent directions from the basal Nubian Sandstone show that uplift in the region of the present Red Sea provided the source area for the sediments (Ward and McDonald, 1979). Later in the Cretaceous, uplift south of Aswan resulted in northward transport and deposition of the main unit of the Nubian Sandstone. Emplacement of alkaline igneous rocks interbedded with, and intruding the sandstones indicates that magmatism postdated the initial doming.

The Tertiary tholeiites south of Quseir are interbedded with the Gebel

Rusas Formation which is interpreted by Johnson (1977) as reflecting the change in direction of sediment transport as the rift valley widened. The small volume of tholeiites may be related to the small amount of crustal separation which has occured (Le Pichon and Francheteau, 1978), reflecting, perhaps, the "vigour" of the thermal plume (Gill, 1974), strength of the lithosphere or northward migration of the African plate, carrying the region from the thermal anomaly. If Harris' (1969) suggestion that the degree of silica saturation correlates with the amount of crustal extension is correct, we may predict that the tholeiites are at most slightly oversaturated.

The tectonic model described above does not address the question of the petrogenesis of the trachytic suite. Both fractional crystallization (e.g. Abbott, 1969; Ferrara and Treuill, 1974) and partial fusion (e.g. McDonald et al., 1970; Bailey, 1978) have been proposed as processes leading to the alkaline basalt-trachyte-peralkaline rhyolite association. While it is beyond the scope of this paper to become involved in this controversy, two observations may be made:

(1) The geographic and temporal proximity of the Wadi Natash lavas and the trachyte intrusions provide circumstantial evidence for a genetic relationship. While there are no basalts associated with the B sites, these may have been removed when the basement complex was uplifted and the overlying sediments removed by erosion. The Wadi Araba lavas, at the northern margin of the basement complex might also be related to the trachytes.

(2) The variations shown on the AFM diagram (Fig. 16) coincide with data from suites interpreted as resulting from both differentiation (Abbot, 1969) and partial fusion (McDonald et al., 1970). Trace element ratios, especially a correlation between Nb and Zr, have been used to support differentiation (Weaver et al., 1972; Ferrara and Treuill, 1974; Gibson, 1974). Within the narrow limits of Zr abundances in the Egyptian trachytes we find no such correlation (Fig. 17). If these elements have not been affected by secondary alteration, out trace element data seem to support partial fusion of more than one source as the origin of the trachytes.

We believe further speculation on the petrogenesis of the alkaline suite is premature without additional data. Such information may be provided by studies in progress of the chemistry of the Wadi Natash lavas (B. Hubbard, pers. commun., 1980) and on the rare earth elements and Sr isotopes of the trachytes. Detailed chemical analyses of the Abu Khrug and Al Khafa ring complexes, where field evidence shows the syenites were preceded by intrusion of essexite gabbros, may also prove relevant (T. Vogel, pers. commun., 1980).

CONCLUSIONS

The study of the minor igneous units formed in the Eastern Desert since the mid Cretaceous shows that they correspond with Falvey's (1974) predictions to a remarkable extent. The paucity of the late phase tholeiitic flows we associate with the low degree of crustal spreading. The chemical studies

of the trachytic rocks are somewhat equivocal with regard to petrogenesis since compositions similar to those reported here have been used in support of both partial fusion and differentiation mechanisms.

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